KAVYA KOLICHELAM

Southfield, MI | +1 (571) 386-9505 | kavyakolichelam@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

MS Computer Science (Data Science) graduate (May 2026, GPA 3.8) with collaborative faculty research in hierarchical deep learning and neuro-symbolic AI, plus 2+ years of applied experience deploying ML models, building BI dashboards, and shipping AI-driven solutions. Built systems benchmarking 10+ deep learning architectures (91.68% classification accuracy, Dice 0.7572 segmentation) and processing 209,527+ records with 1.4% Expected Calibration Error. Skilled in Python, SQL, PyTorch, TensorFlow, scikit-learn, and end-to-end ML workflows; hands-on with Generative AI (LLMs, RAG), prompt engineering, deep learning, computer vision, and NLP. Proven ability to translate complex analytical findings into actionable insights for technical and non-technical stakeholders across cross-functional teams.

TECHNICAL SKILLS

Languages & Querying: Python, SQL, Java, C, C++

Machine Learning: Regression, Classification, Ensemble Methods (XGBoost, Random Forest), Feature Engineering, Model Evaluation, Cross-Validation, Hyperparameter Tuning, Scikit-learn, Probabilistic ILP

Deep Learning & CV: PyTorch, TensorFlow, Keras, CNN, MobileNetV2, SegFormer, U-Net++, Transfer Learning, Domain Adaptation, Multi-Task Learning, Fine-Tuning

NLP & Generative AI: TF-IDF, Sentiment Analysis, NLTK, spaCy, HuggingFace Transformers, DistilBERT, LLMs, RAG, Prompt Engineering, OpenAI APIs, Vector Retrieval

Data Analysis & Statistics: EDA, Statistical Analysis, Hypothesis Testing, A/B Testing, Predictive Analytics, Trend Analysis, Calibration Analysis (ECE), Data Mining

Data Processing: Pandas, NumPy, SciPy, Data Cleaning, Data Validation, Data Wrangling, ETL Concepts, Feature Engineering

Visualization & BI: Power BI, Tableau, Matplotlib, Seaborn, NetworkX, WordCloud, Excel (Pivot Tables, VLOOKUP, XLOOKUP, Power Query), Google Sheets

Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code, Google Colab, MS Office Suite, Streamlit

PROFESSIONAL EXPERIENCE

Teaching Assistant – Machine Learning | Lawrence Technological University, Southfield, MI *Aug 2025 – May 2026*

- Mentored 40+ graduate students across 2 academic semesters in ML model development, data preprocessing, feature engineering, and predictive modeling for course assignments and capstone projects
- Reviewed and validated 100+ student-built ML pipelines; diagnosed processing errors and guided systematic corrections that improved assignment result accuracy and student confidence with end-to-end workflows
- Translated complex ML and statistical concepts — including regression, classification, model evaluation, calibration, and hyperparameter tuning — into clear explanations for students with diverse technical backgrounds
- Supported technical documentation, grading, and cross-team coordination across multiple course deliverables, ensuring consistent feedback turnaround against tight semester deadlines

Machine Learning Engineer | Start Tech Innovations, India *Jan 2023 – Dec 2023*

- Developed and deployed regression and classification models (Python, Scikit-learn, XGBoost) into production business workflows across multiple structured datasets, enabling data-driven decisions across the organization
- Conducted comprehensive EDA, feature engineering, and statistical analysis — surfacing trends, anomalies, and optimization opportunities that informed model design and improved baseline accuracy through systematic iteration
- Evaluated models rigorously using cross-validation, precision, recall, ROC-AUC, and RMSE; iterated on hyperparameters, feature selection, and class-balance strategies to reach production-quality performance
- Designed and automated scalable ML data pipelines, eliminating manual preprocessing steps and ensuring reliable, repeatable outputs for recurring reporting cycles
- Translated analytical findings into actionable recommendations for business stakeholders across cross-functional teams, driving data-informed operational decisions

Data Analyst Intern | Tech Solutions Pvt Ltd, Hyderabad, India *May 2022 – Dec 2022*

- Analyzed large structured datasets using SQL and Python (Pandas, NumPy) to extract insights, identify trends, and support data-driven decision-making across multiple departments

- Performed statistical analysis, data mining, and predictive modeling to uncover optimization opportunities and surface data quality issues, improving downstream reporting accuracy
- Built and maintained Power BI dashboards and automated reports enabling self-service analytics for non-technical stakeholders, reducing manual reporting effort and accelerating decision cycles
- Cleaned, validated, and reprocessed Excel and CSV datasets; investigated discrepancies and implemented systematic fixes that improved downstream KPI tracking accuracy
- Collaborated with cross-functional teams to deliver insights supporting operational efficiency, KPI performance, and recurring business reporting requirements

PROJECT EXPERIENCE

Neuro-Symbolic Hybrid Classification System | *Python*, *Scikit-learn*, *TF-IDF*, *Probabilistic ILP*, *Multinomial Naive Bayes*, *NetworkX*

- Designed a two-tier hybrid AI architecture combining Probabilistic Inductive Logic Programming (P-ILP) rules with calibrated Multinomial Naive Bayes for interpretable text classification on 209,527 HuffPost news articles across 10 categories
- Achieved 82.85% overall accuracy with 24.3% of decisions made through transparent rule-based logic and 75.7% through calibrated ML; Expected Calibration Error of 1.4% (target <5%) validates confidence score reliability
- Implemented and compared four fusion strategies (Confidence Routing, Weighted Voting, Cascaded Fusion, Stacking Ensemble); achieved 91.3% rule-ML fidelity confirming ante-hoc interpretability
- Documented as Collaborative Research Project formal report covering literature review, methodology, hypothesis validation, and limitations

Social Network Sentiment Analysis for Mental Health Awareness | *Python*, *NLP*, *HuggingFace Transformers*, *DistilBERT*, *NLTK*, *Scikit-learn*

- Built end-to-end NLP pipeline classifying sentiment across 38,033 Reddit posts using VADER for three-class rule-based labeling and DistilBERT for transformer-based binary classification
- Applied K-means thematic clustering (k=5) on TF-IDF features with PCA dimensionality reduction; combined with WordCloud and temporal trend analysis to surface latent themes for stakeholder reporting
- Co-authored project: handled imbalanced classes, produced clear visualizations using Matplotlib and Seaborn, and communicated findings to non-technical audiences through structured visual reports

Hierarchical Coarse-to-Fine Crack Detection Framework | **Collaborative Research Project with Dr. Wisam Bukaita** | *PyTorch*, *SegFormer-B2*, *MobileNetV2*, *U-Net++*, *Transfer Learning*, *Domain Adaptation*

- Designed and implemented a three-stage deep learning framework for automated concrete structural inspection: MobileNetV2 classifier (91.68% accuracy, ROC-AUC 0.934, Macro-F1 0.844), SegFormer-B2 segmentation at 384×384 with TTA (Dice 0.7572, IoU 0.6353), and geometry stage estimating crack length and width from masks
- Benchmarked 10+ deep learning architectures including U-Net variants, Attention U-Net, DeepLabV3+, U-Net++ ResNet34, EfficientNet-B4 U-Net++, and SegFormer-B0/B2 before selecting SegFormer-B2 — improving segmentation Dice by 0.08+ over baseline U-Net
- Implemented two-stage domain adaptation on the DeepCrack dataset, improving cross-domain Dice to 0.6553; designed pipeline-level reliability metrics filtering 85.06% of images before segmentation (reducing inference cost significantly)
- Conducted statistical validation, ablation studies, and computational efficiency analysis across 56K+ images; documented results in formal research manuscript

LLM-Based AI Assistant with Retrieval-Augmented Generation | *OpenAI APIs*, *RAG*, *Prompt Engineering*, *Python*, *Vector Retrieval*

- Designed and developed a Retrieval-Augmented Generation (RAG) chatbot using OpenAI APIs, implementing structured document indexing, semantic retrieval, and context-aware prompt engineering
- Integrated dynamic retrieval pipeline grounding LLM responses in relevant source documents, improving factual accuracy and reducing hallucinations
- Applied advanced prompt engineering techniques including chain-of-thought prompting and few-shot examples to optimize response quality across diverse query types

EDUCATION

Master of Science – Computer Science (Data Science) GPA: 3.8 / 4.0

Lawrence Technological University, Southfield, MI *May 2026*

Relevant Coursework: *Machine Learning*, *Deep Learning*, *Natural Language Processing*, *Statistical Learning*, *Data Mining*, *Algorithms & Data Structures*, *Probability & Statistics*

Bachelor of Technology – Electrical & Electronics Engineering

Bhoj Reddy Engineering College for Women, Hyderabad, India *2019 – 2023*

CERTIFICATIONS

- Machine Learning – Coursera (Andrew Ng)
- Deep Learning Specialization – Coursera
- Generative AI Fundamentals – Coursera
- Data Science Tools – IBM
- Python for Data Science – IBM
- SQL for Data Science – Coursera
- Data Analysis with Python – IBM / Coursera
- Data Visualization with Python – IBM / Coursera
- Excel Basics for Data Analysis – IBM / Coursera
- Power BI for Beginners – Microsoft / Coursera